

Learning Roadmap

1. Understand MOSFETs
2. Basic Buck principles
3. Buck design
4. LTspice simulation
5. PCB Layout for switching

MOSFETs

- Apply voltage gate pin to allow current to flow from drain-source * depends on P or N channel

* How does it work without current?

- Gate is physically blocked
 - ↳ ΔV across it creates E-field which induces I flow from Drain-source
- FET holds charge from being turned on
 - ↳ Need to connect Gate to GND to turn off
 - ↳ Use resistors to minimize inrush and discharge current

Enhancement: normally off. V applied to gate turns on

Depletion: normally on. V applied to gate turns off

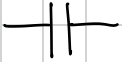
* look for broken/solid line in symbol (enhancement/depletion)

N-Type: Voltage applied to drain

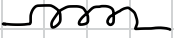
P-Type: Voltage applied to source

- Physical channel forms in FET for
 - ↳ Depletion type already has a channel

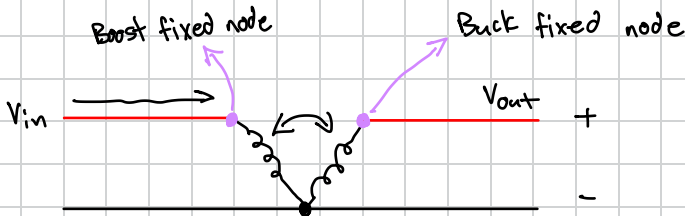
DC-DC Converter Basics



- Cap maintains ΔV across terminals
↳ Produces current to do this



- Inductor maintains current flow through it
↳ Acts as V source to do this



- Since inductor keeps I flow direction, V_{out} flows in not out
↳ Negative V w/ respect to V_{in}

How to switch inductor position?

- Use two switches to direct the path of current
↳ Can replace one switch with diode for discharge path

* 50% duty for buck-boost $-V_{in}$

Buck: $V_{out} = V_{in} \cdot d$

Boost: $V_{out} = V_{in} \cdot \frac{1}{1-d}$

Buck Boost: $V_{out} = V_{in} \cdot \frac{-d}{1-d}$

DC-DC Converter Design

- Real converters don't match 100% w/ simulation/calculation
↳ Power conservation gives rough idea of V and I
- At lower output current, inductor current goes to zero
↳ L transfers excess energy to C causing higher voltage instead of just smoothing
↳ $V_{out} \uparrow$

Calculating Inductance

$$V = L \frac{di}{dt}$$

$$L = \frac{V dt}{di}$$

$$L = \frac{\int V}{\Delta I}$$

Switch Closed: V positive $\therefore di/dt$ positive $\therefore$ current $\uparrow$

Switch Open: V negative $\therefore di/dt$ negative $\therefore$ current $\downarrow$

* ΔI should be calculated base on minimum expected load

2x avg. of I_L waveform



Switch States

Buck Phase 1 : Charges inductor

Buck Phase 2 : Creates loop with L as only V source

Boost Phase 1 : Charges L , loops I without reachin V_{out}

Boost Phase 2 : Discharges V_{out} and V_L to V_{out}

Buck-Boost Phase I : Creates loop without V_{out}
BB 2 : Creates negative loop w/ V_{out}

$$I = C \frac{dV}{dt}$$

$$C = I \frac{dt}{dV}$$

$$C = \frac{\int I}{\Delta V}$$

$$P = V \cdot I$$

$$I = P/V$$

Key Design Considerations

- Switching frequency
- Control type (open/closed)
- Power rating
 - ↳ Synchronous?
- Peak current vs. Voltage control mode
- Type I or Type II control?
- MOSFET type (N vs. P channel)

Basic Design Specs

24 V - from what source?

- 6S LiPo battery has broad applications

Voltage range : 4.2V/cell - 3.0V/cell

* 25.2V - 18.0V

What am I powering?

- Most of my projects will use MCU's

* 100 mA - 500 mA

$$V_{in} = 25.2 \text{ V}$$

$$V_{out} = 5 \text{ V}$$

$$D = V_{out} / V_{in} \approx 0.198 \text{ (19.8\%)} \quad \text{MIN}$$

$$V_{in} = 18.0 \text{ V} \quad D = V_{out} / V_{in} \approx 0.278 \text{ (27.8\%)} \quad \text{MAX}$$

$$V_{out} = 5.0 \text{ V}$$

@ 500 mA output, 85% efficiency

$$P_{out} = 5 \text{ V} \cdot 500 \text{ mA} = 2.5 \text{ W}$$

$$P_{in} \times 0.85 = P_{out} \quad P_{in} \approx 2.94 \text{ W}$$

$$P_{loss} \approx 0.441 \text{ W} \quad \text{MAX}$$

@ 100 mA output 85% efficiency

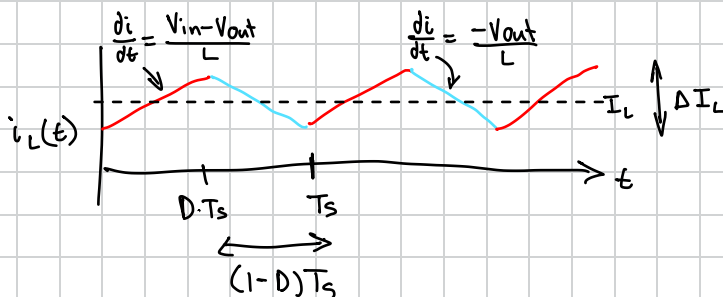
$$P_{loss} \approx 0.088 \text{ W (88 mW)} \quad \text{MIN}$$

Lower $f_{sw} = \downarrow P_{loss}$

- MOSFET loses power on every open/close
- Gate driver must be able to produce $t_{on} = \frac{D}{f_{sw}}$

Switching Frequency & Inductance

V_L avg must equal 0 or else energy will accumulate



$$T_s = 1 / f_{sw}$$

* Current through inductor is equal to current through load

- To control 100mA - 500mA, adjust I_L
↳ How?

↙ Target

$$I_R = 500 \text{ mA} = I_L = V_o / R$$

$$I_{L\max} = I_L + \frac{\Delta i_L}{2}$$

$$V = L \frac{di}{dt} \quad \frac{V}{L} = \frac{di}{dt}$$

$$\Delta i_L = \text{slope} \cdot \text{time}$$
$$\Delta i_L = \frac{V_{\text{out}}}{L} \cdot (1-D) T_S$$

↙
 $1/f_{\text{sw}}$

For CCM $\Delta i_L < 2 \times I_L$

* $I_{L\min} = 100 \text{ mA} \quad I_{L\max} = 500 \text{ mA}$

- Must choose f_{sw} that works for both I_L values

$$\frac{5V(1-0.278)}{L \cdot f_{\text{sw}}} \leq 2 \cdot 100 \text{ mA} \quad L \cdot f_{\text{sw}} > 18.05$$

$$\frac{5V(1-0.198)}{L \cdot f_{\text{sw}}} \leq 2 \cdot 500 \text{ mA} \quad L \cdot f_{\text{sw}} \geq 4.08$$

$$L \cdot f_{\text{sw}} \geq 18.05$$

$$L = \frac{\int V}{\Delta I}$$

* ΔI_L 30% - 40 %
of $I_{\max}$ for
standard practice

Pick f_{sw} **BEFORE** finding inductance

How to choose f_{sw}

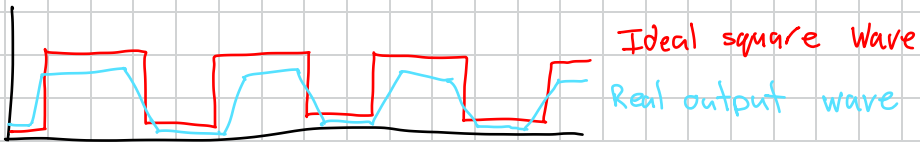
Design considerations:

- Control
 - ↳ Dedicated IC?
 - ↳ Custom control loop? ✓
 - ↳ MCU

How do analog components inform f_{sw} ?

- Slew rate
- Propagation delay

Slew Rate: $\Delta V/t$ i.e. ability to change output signal by a certain amount in a given time
max slope of op amp's output



Propagation Delay: Time between input and output wave generation

$$t_{pd} \ll t_{on}$$

- For this project let $t_{pd} \leq \frac{t_{on}}{20}$ * (5% of ON period)

@ $f_{sw} = 100 \text{ kHz}$ $t_{on} = 10 \mu\text{s}$ $t_{pd} \leq 0.5 \mu\text{s} / 500 \text{ ns}$

* Ample supply of affordable comparators with low t_{pd}

* $f_{sw} = 100 \text{ kHz}$

Why?

- More modern silicon tech for analog components
- Simple round number for calculations
- Allows for smaller L
 - ↳ Can increase to $\sim 200 \text{ kHz}$ and still have $t_{pd} < 5\%$
- Minimize power losses

Inductor Sizing

ripple ratio = 0.35

max load = 500 mA $\Delta i_L = 175 \text{ mA}$

$$\Delta i_L = \frac{5V(1-0.198)}{L \cdot f_{sw}}$$

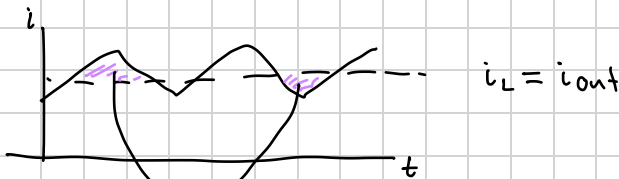
$$L = \frac{5V(1-0.198)}{0.175 \text{ A} \cdot 100 \text{ kHz}} \approx 0.229 \text{ mH}$$

Standard should actually be 30%

Closest standard value: $220 \mu\text{H}$

$$f_{sw} = \frac{5V(1-0.198)}{(220 \times 10^{-6})(150 \times 10^3)} = 121 \text{ kHz}$$

Capacitor Sizing



equal areas
in steady
state

$$C = \frac{T \Delta i_L}{8 \Delta V_O}$$

$$A = \frac{1}{2} b \cdot h$$

$$b = t_{on}$$

$$h = \Delta i_L / 2$$

* ΔV_O should be
1% - 2% of V_{out}

$$Q = \frac{1}{4} t_{on} \cdot \Delta i_L = 1.54 \times 10^{-10} \text{ C}$$

ΔV_{out} Method:

$$\text{let } \Delta V_{out} = 0.01 (5V) = 0.05V$$

$$\Delta V_{out} = \Delta V_C + \Delta V_{ECR} \quad \text{let } \Delta V_C = \Delta V_{ECR} = 25mV$$

$$C = \frac{T_{\Delta i_L}}{8 \Delta V_C} = \frac{\Delta i_L}{8 \cdot f_{sw} \cdot \Delta V_C} = \frac{150mA}{8 \cdot 121kHz \cdot 25mV} = 6nF$$

$$ESR = \frac{0.025V}{0.15A} = 167m\Omega$$

Transient Response Method:

$$C_{out} = \frac{L \cdot (\Delta I)^2}{2 V_{out} \Delta V_{trans}}$$

$$\text{let } \Delta V_{trans} = V_{out} \cdot 0.03 = 0.15V$$

$$\Delta I = \text{max swing } 500mA - 100mA = 400mA$$

$$C_{out} = \frac{(220\mu H)(400mA)^2}{2(5V)(0.15V)} \approx 23.47\mu F$$

* 22 μF nearest standard value

Key Parameters

$$f_{sw} = 121kHz$$

$$L = 220\mu H$$

$$C_{out} = 22\mu F \quad ESR \leq 167m\Omega$$

$$T_{pd} \leq 413ns \quad T_{rise} \leq 165ns$$

Buck Topology Choice

Synchronous vs. Asynchronous

let V_f of diode equal $0.3V$

$$I_{out} = 500 \text{ mA} \quad P_{loss} = (0.3V) \cdot (500 \text{ mA}) (1 - 0.198) \approx 120 \text{ mW}$$

$$P_{total} = (5V) \cdot (500 \text{ mA}) = 2.5 \text{ W}$$

$$\text{Eff}_{target} = 0.85 \quad \text{Allowable losses} = 0.15 \times 2.5 \text{ W} = 375 \text{ mW}$$

$120 \text{ mW} \ll 375 \text{ mW} \therefore$ asynchronous topology is viable

Diode Selection

- Schottky diodes have lowest V_f (important for efficiency)
 - $\hookrightarrow 0.2V - 0.4V$
- V rating 20% - 50% greater than max V_{in}
 - $\hookrightarrow 30V - 40V$ approx.
- I rating of 1A - 2A
 - $\hookrightarrow$ Account for I ripple and transient spikes

MOSFET Selection

- Enhancement type
 - $\hookrightarrow$ N-channel or P-channel?
- High side switching

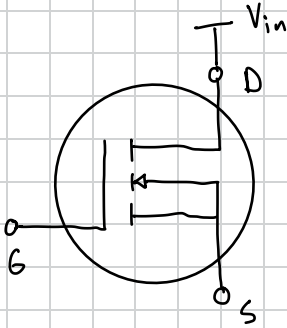
N-channel FETs have $\downarrow R_{DS(on)}$

- e^- used to bridge source and drain
 - $\hookrightarrow e^-$ "move" faster than holes
- More efficient + faster switching

High Side N-channel Drive

Current flows drain $\rightarrow$ source

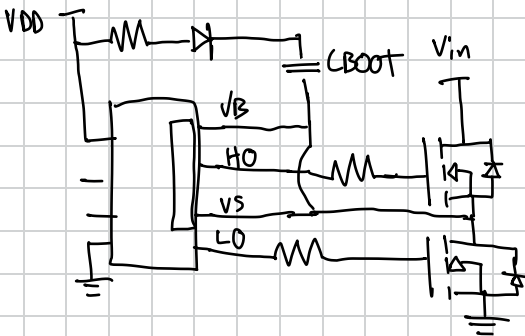
$V_{in}(25.2V - 18V)$



Gate driver must be able to supply $> 25.2V$

$$V_{GS} > V_{DS}$$

Bootstrap Drive



- When high side FET is open and low side is closed
I flows through C_{BOOT}
 $\rightarrow V_S < V_{DD}$
- $V_{BOOT} \uparrow$ to V_{DD} value
 $\rightarrow D_{BOOT}$ blocks current flow
 $\rightarrow$ I flows from V_S , into V_B and out of H_0
 $\rightarrow V_B = V_{GS}$ to switch FET

Input Capacitor

C_{in} gets charged when high-side FET is off

$$C_{IN} \geq \frac{D \times (1-D) \times I_o}{\Delta V_{in} \times f_{sw}}$$

$$I_{max} = 500 \text{ mA} \quad \text{let } \Delta V_{in} = 1 \text{ mV}$$

$$f_{sw} = 121 \text{ kHz}$$

• For worst case calc $D \times (1-D)$ greatest at 27.8%

$$C_{IN} \geq \frac{0.278(1-0.278) \times 500 \text{ mA}}{1 \text{ mV} \times 121 \text{ kHz}} = 829.4 \text{ nC}$$